

LANDSCAPES OF THE CHURCH

00 — Land-Cover Composition

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

LSAP Integral Ecology Atlas | Technical Documentation | 3 October 2026

5. Analysis 00 — Land-Cover Composition

5.1 Diocesan layer overview

Layer name	LandscapesChurch_00_Dioceses_Composition_V2
Layer alias	00 - Diocesan Land-Cover Composition
Scale / features	Diocese; 2,874 features
Description	Foundational diocesan land-cover composition layer showing the percentage and area of the eight valid land-cover systems across each Catholic diocesan analytical geography. It provides the baseline from which the thematic 01–06 derivatives are calculated.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Eight-class composition; class areas; coverage/context; Church storytelling; separate water-context note.
Key story fields	STORY_LAND; STORY_MOSAIC; STORY_SCALE; STORY_CHURCH; STORY_CONTEXT; WATER_NOTE
Status	Publication-ready

5.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_00_Conferences_Composition_V2
Layer alias	00 - Episcopal Conference Land-Cover Composition
Scale / features	Episcopal Conference; 122 features
Description	Foundational Episcopal Conference land-cover composition layer showing the percentage and area of the eight valid land-cover systems across each Conference analytical geography, paired with Church and LSAP context.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).

Analyses	Eight-class composition; class areas; coverage/context; Conference leadership and LSAP storytelling; separate water-context note.
Key story fields	STORY_LAND; STORY_CHURCH; STORY_LSAP; STORY_ACTION; STORY_INTEGRAL; STORY_SUMMARY; WATER_NOTE
Status	Publication-ready

5.3 Analytical workflow

1. Copy the diocesan or Conference biodiversity-summary source into a dedicated Analysis 00 derivative feature class.
2. Verify the eight valid PCT_LULC_* and source TOTAL_LULC_* fields.
3. Diagnose ANALYSIS_AREA / area fields before conversion; convert m² to km² exactly once.
4. Retain Water as part of full landscape composition but keep water interpretation out of general ecclesiastical-jurisdiction prose.
5. Generate scale-appropriate Church/LSAP stories.
6. Run completeness QC for all story fields and formula/area plausibility checks.
7. Remove unrelated biodiversity, carbon, terrain, and population fields from the publication derivative while preserving source/master data.

5.4 Core QC retained from workflow

Retained QC output

```
LANDSCAPES OF THE CHURCH – V2
00 – LAND-COVER COMPOSITION
CONFERENCE STORY WRITE / COMPLETENESS
```

```
1. INPUT VERIFICATION
Features: 122
Required fields: PASS
```

```
2. CURSOR VERIFICATION
Cursor fields: 31
STORY_LAND index: 25
STORY_CHURCH index: 26
STORY_LSAP index: 27
STORY_ACTION index: 28
STORY_INTEGRAL index: 29
STORY_SUMMARY index: 30
```

```
4. BASIC COMPLETENESS QC
STORY_LAND missing=0
STORY_CHURCH missing=0
STORY_LSAP missing=0
STORY_ACTION missing=0
STORY_INTEGRAL missing=0
STORY_SUMMARY missing=0
```

```
Additional retained area/formula QC:
Conference records: 122
Median area agreement ratio: 0.999986
Negative / >105% area cases: 0
Formula errors: 0
```

Relevant reproducible code

A.1 Hexagon intersection and overlap-area-weighted summaries

```
import arcpy
import os
from collections import defaultdict

GDB = r"C:\path\to\project.gdb"

TARGET = os.path.join(GDB, "Target_Geography")      # dioceses, conferences, or buffers
HEX = os.path.join(GDB, "BiodiversityHexagons_L5")
INTERSECT = os.path.join(GDB, "Target_Hex_Intersect")

TARGET_ID = "TARGET_ID"
HEX_ID = "GRID_ID"

# Example rate/mean fields.
WEIGHTED_FIELDS = [
    "PCT_LULC_Trees",
    "PCT_LULC_Crops",
    "PCT_LULC_Built",
    "PCT_PROTECTED_AREA",
    "Biodiversity_Intactness",
    "Tree_Canopy",
    "Carbon_Density",
    "Slope_Mean",
    "Elevation_Mean"
]

arcpy.analysis.Intersect(
    [TARGET, HEX],
    INTERSECT,
    join_attributes="ALL"
)

# Use geodesic area when working globally.
if "OVERLAP_KM2" not in {f.name for f in arcpy.ListFields(INTERSECT)}:
    arcpy.management.AddField(INTERSECT, "OVERLAP_KM2", "DOUBLE")

arcpy.management.CalculateGeometryAttributes(
    INTERSECT,
    [["OVERLAP_KM2", "AREA_GEODESIC"]],
    area_unit="SQUARE_KILOMETERS"
)

# Numerator and denominator are maintained separately for EACH field so nulls
# do not dilute the denominator.
num = defaultdict(lambda: defaultdict(float))
den = defaultdict(lambda: defaultdict(float))

fields = [TARGET_ID, "OVERLAP_KM2"] + WEIGHTED_FIELDS

with arcpy.da.SearchCursor(INTERSECT, fields) as cur:
    for row in cur:
        target_id = row[0]
        overlap = float(row[1] or 0)

        if overlap <= 0:
            continue

        for i, fld in enumerate(WEIGHTED_FIELDS, start=2):
            value = row[i]

            if value is None:
                continue

            num[target_id][fld] += float(value) * overlap
            den[target_id][fld] += overlap

weighted = defaultdict(dict)
```

```

for target_id in num:
    for fld in WEIGHTED_FIELDS:
        if den[target_id][fld] > 0:
            weighted[target_id][fld] = (
                num[target_id][fld] /
                den[target_id][fld]
            )
        else:
            weighted[target_id][fld] = None

# Join/write these weighted values back to a copy of TARGET using TARGET_ID.

```

A.3 Shared land-cover helpers

```

import math

LANDCOVER = {
    "Water": "PCT_LULC_Water",
    "Trees": "PCT_LULC_Trees",
    "Cropland": "PCT_LULC_Crops",
    "Built": "PCT_LULC_Built",
    "Rangeland": "PCT_LULC_Rangeland",
    "Bare": "PCT_LULC_Bare",
    "Snow/Ice": "PCT_LULC_SnowIce",
    "Other Vegetation": "PCT_LULC_Veg"
}

def safe(v):
    return 0.0 if v is None else float(v)

def normalized_classified(raw):
    valid = sum(max(0.0, safe(v)) for v in raw.values())

    if valid <= 0:
        return valid, {k: 0.0 for k in raw}

    return (
        valid,
        {k: 100.0 * max(0.0, safe(v)) / valid for k, v in raw.items()}
    )

def shannon_metrics(raw):
    valid, pct = normalized_classified(raw)

    props = {k: v / 100.0 for k, v in pct.items()}

    H = -sum(
        p * math.log(p)
        for p in props.values()
        if p > 0
    )

    diversity = 100.0 * H / math.log(8)
    effective = math.exp(H)

    meaningful = sum(v >= 1.0 for v in pct.values())
    major = sum(v >= 5.0 for v in pct.values())

    ordered = sorted(
        raw.items(),
        key=lambda kv: safe(kv[1]),
        reverse=True
    )

    top_type, top_pct = ordered[0]
    second_type, second_pct = ordered[1]

    classified_gap = (
        pct[top_type] -
        pct[second_type]
    )

    return {

```

```

        "valid_cover": valid,
        "diversity": diversity,
        "effective": effective,
        "meaningful": meaningful,
        "major": major,
        "top_type": top_type,
        "top_pct": safe(top_pct),
        "second_type": second_type,
        "second_pct": safe(second_pct),
        "classified_gap": classified_gap
    }

def mosaic_context(m):
    if (
        m["diversity"] >= 65 and
        m["effective"] >= 4.0 and
        m["meaningful"] >= 5 and
        m["major"] >= 4 and
        m["classified_gap"] <= 15
    ):
        return "Highly Diverse Mosaic"

    if (
        m["diversity"] >= 55 and
        m["effective"] >= 3.25 and
        m["meaningful"] >= 4 and
        m["major"] >= 3 and
        m["classified_gap"] <= 35
    ):
        return "Diverse Mosaic"

    if (
        m["diversity"] < 35 or
        m["effective"] < 2.0 or
        m["major"] <= 1 or
        m["classified_gap"] >= 60
    ):
        return "Concentrated Landscape"

    return "Structured Mosaic"

```

A.10 Safe m² → km² conversion

```

import statistics

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")

```

A.16 Publication cleanup pattern

Keep only fields intentionally required for the publication derivative.

```
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
    "Official_website",
    "Sector",
    "BUFFER_AREA",
    "LANDSCAPE_SCALE",
    # ... eight PCT_LULC fields ...
    # ... coverage/mosaic/transition fields ...
    # ... story fields ...
}
```

```
delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]
```

```
if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )
```

Always run a final required-field check after deletion.

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbdcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>